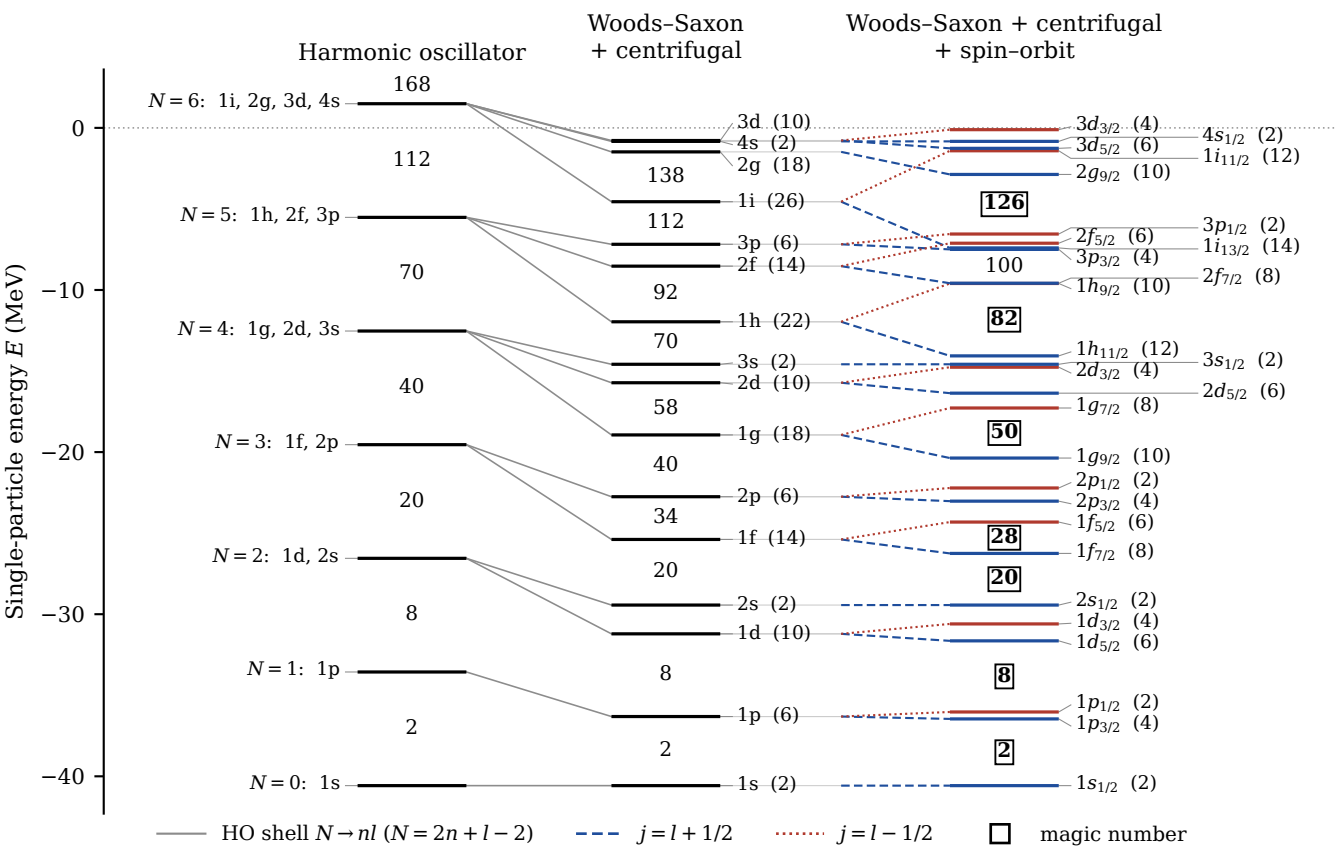
Harmonic oscillator
Woods–Saxon + centrifugal
Woods–Saxon + centrifugal + spin–orbit
Single-particle energy E (MeV)
0
−10
−20
−30
−40
$N = 6$: 1i, 2g, 3d, 4s
$N = 5$: 1h, 2f, 3p
$N = 4$: 1g, 2d, 3s
$N = 3$: 1f, 2p
$N = 2$: 1d, 2s
$N = 1$: 1p
$N = 0$: 1s
168
112
70
40
20
8
2
138
112
92
70
58
40
34
20
8
2
3d (10)
4s (2)
2g (18)
1i (26)
3p (6)
2f (14)
1h (22)
3s (2)
2d (10)
1g (18)
2p (6)
1f (14)
2s (2)
1d (10)
1p (6)
1s (2)
$3d_{3/2}$ (4)
$3d_{5/2}$ (6)
$2g_{9/2}$ (10)
126
$2f_{5/2}$ (6)
$3p_{3/2}$ (4)
$1h_{9/2}$ (10)
82
$1h_{11/2}$ (12)
$2d_{3/2}$ (4)
$1g_{7/2}$ (8)
50
$1g_{9/2}$ (10)
$2p_{1/2}$ (2)
$2p_{3/2}$ (4)
$1f_{5/2}$ (6)
28
$1f_{7/2}$ (8)
20
$2s_{1/2}$ (2)
$1d_{3/2}$ (4)
$1d_{5/2}$ (6)
8
$1p_{1/2}$ (2)
$1p_{3/2}$ (4)
2
$1s_{1/2}$ (2)
$4s_{1/2}$ (2)
$1i_{11/2}$ (12)
$3p_{1/2}$ (2)
$1i_{13/2}$ (14)
$2f_{7/2}$ (8)
$3s_{1/2}$ (2)
$2d_{5/2}$ (6)
HO shell $N \to nl$ ($N = 2n + l - 2$)
$j = l + 1/2$
$j = l - 1/2$
magic number